

Learning to Be Me by Greg Egan

I was six years old when my parents told me that there was a small, dark jewel inside my skull, learning to be me.

Microscopic spiders had woven a fine golden web through my brain, so that the jewel's teacher could listen to the whisper of my thoughts. The jewel itself eavesdropped on my senses, and read the chemical messages carried in my bloodstream; it saw, heard, smelt, tasted and felt the world exactly as I did, while the teacher monitored its thoughts and compared them with my own. Whenever the jewel's thoughts were *wrong*, the teacher-faster than thought-rebuilt the jewel slightly, altering it this way and that, seeking out the changes that would make its thoughts correct.

Why? So that when I could no longer be me, the jewel could do it for me.

I thought: if hearing that makes *me* feel strange and giddy, how must it make the *jewel* feel? Exactly the same, I reasoned; it doesn't know it's the jewel, and it too wonders how the jewel must feel, it too reasons: "Exactly the same; it doesn't know it's the jewel, and it too wonders how the jewel must feel..."

And it too wonders-

(I knew, because *I* wondered)

-it too wonders whether it's the real me, or whether in fact it's only the jewel that's learning to be me.

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As a scornful twelve-year-old, I would have mocked such childish concerns. Everybody had the jewel, save the members of obscure religious sects, and dwelling upon the strangeness of it struck me as unbearably pretentious. The jewel was the jewel, a mundane fact of life, as ordinary as excrement. My friends and I told bad jokes about it, the same way we told

bad jokes about sex, to prove to each other how blase we were about the whole idea.

Yet we weren't quite as jaded and imperturbable as we pretended to be. one day when we were all loitering in the park, up to nothing in particular, one of the gang-whose name I've forgotten, but who has stuck in my mind as always being far too clever for his own good-asked each of us in turn: "Who *are* you? The jewel, or the real human?" We all replied-unthinkingly, indignantly-"The real human!" When the last of us had answered, he cackled and said, "Well, I'm not. *I'm* the jewel. So you can eat my shit, you losers, because *you'll* all get flushed down the cosmic toilet-but me, I'm gonna live forever."

We beat him until he bled.

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By the time I was fourteen, despite-or perhaps because of-the fact that the jewel was scarcely mentioned in my teaching machine's dull curriculum, I'd given the question a great deal more thought. The pedantically correct answer when asked "Are you the jewel or the human?" had to be "The human"-because only the human brain was physically able to reply. The jewel received input from the senses, but had no control over the body, and its intended reply coincided with what was actually said only because the device was a perfect imitation of the brain. To tell the outside world "I am the jewel"-with speech, with writing, or with any other method involving the body-was patently false (although to *think it* to oneself was not ruled out by this line of reasoning).

However, in a broader sense, I decided that the question was simply misguided. So long as the jewel and the human brain shared the same sensory input, and so long as the teacher kept their thoughts in perfect

step, there was only *one* person, *one* identity, *one* consciousness. This one person merely happened to have the (highly desirable) property that *if either* the jewel *or* the human brain were to be destroyed, he or she would survive

unimpaired. People had always had two lungs and two kidneys, and for almost a century, many had lived with two hearts. This was the same: a matter of redundancy; a matter of robustness, no more.

That was the year that my parents decided I was mature enough to be told that they had both undergone the switch-three years before. I pretended to take the news calmly, but I hated them passionately for not having told me at the time. They had disguised their stay in hospital with lies about a business trip overseas. For three years I had been living with jewel-heads, and they hadn't even told me. It was *exactly* what I would have expected of them.

“We didn't seem any different to you, did we?” asked my mother.

“No,” I said-truthfully, but burning with resentment nonetheless.

“That's why we didn't tell you,” said my father. “If you'd known we'd switched, at the time, you might have *imagined* that we'd changed in some way. By waiting until now to tell you, we've made it easier for you to convince yourself that we're still the same people we've always been.” He put an arm around me and squeezed me. I almost screamed out, “Don't *touch* me!” but I remembered in time that I'd convinced myself that the jewel was No Big Deal.

I should have guessed that they'd done it, long before they confessed; after all, I'd known for years that most people underwent the switch in their early thirties. By then, it's downhill for the organic brain, and it would be foolish to have the jewel mimic this decline. So, the nervous system is rewired; the reins of the body are handed over to the jewel, and the teacher is deactivated. For a week, the outward-bound impulses from the brain are compared with those from the jewel, but by this time the jewel is a perfect copy, and no differences are ever detected.

The brain is removed, discarded, and replaced with a spongy tissue-cultured object, brain-shaped down to the level of the finest capillaries, but no more capable of thought than a lung or a kidney. This mock-brain removes exactly as much oxygen and glucose from the blood as the real thing, and

faithfully performs a number of crude, essential biochemical functions. In time, like all flesh, it will perish and need to be replaced.

The jewel, however, is immortal. Short of being dropped into a nuclear fireball, it will endure for a billion years.

My parents were machines. My parents were gods. It was nothing special. I hated them.

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When I was sixteen, I fell in love, and became a child again.

Spending warm nights on the beach with Eva, I couldn't believe that a mere machine could ever feel the way I did. I knew full well that if my jewel had been given control of my body, it would have spoken the very same words as I had, and executed with equal tenderness and clumsiness my every awkward caress-but I couldn't accept that its inner life was as rich, as miraculous, as joyful as mine. Sex, however pleasant, I could accept as a purely mechanical function, but there was something between us (or so I believed) that had nothing to do with lust, nothing to do with words, nothing to do with any tangible action of our bodies that some spy in the sand dunes with parabolic microphone and infrared binoculars might have discerned. After we made love, we'd gaze up in silence at the handful of visible stars, our souls conjoined in a secret place that no crystalline computer could hope to reach in a billion years of

striving. (If I'd said *that* to my sensible, smutty, twelve-year-old self, he would have laughed until he haemorrhaged.)

I knew by then that the jewel's "teacher" didn't monitor every single neuron in the brain. That would have been impractical, both in terms of handling the data, and because of the sheer physical intrusion into the tissue. Someone-or-other's theorem said that sampling certain critical neurons was almost as good as sampling the lot, and-given some very reasonable

assumptions that nobody could disprove-bounds on the errors involved could be established with mathematical rigour.

At first, I declared that *within these errors*, however small, lay the difference between brain and jewel, between human and machine, between love and its imitation. Eva, however, soon pointed out that it was absurd to make a radical, qualitative distinction on the basis of the sampling density; if the next model teacher sampled more neurons and halved the error rate, would its jewel then be “half-way” between

“human” and “machine?” In theory-and eventually, in practice-the error rate could be made smaller than any number I cared to name. Did I really believe that a discrepancy of one in a billion made any difference at all-when every human being was permanently losing thousands of neurons every day, by natural attrition?

She was right, of course, but I soon found another, more plausible, defence for my position. Living neurons, I argued, had far more internal structure than the crude optical switches that served the same function in the jewel's so-called “neural net.” That neurons fired or did not fire reflected only one level of their behaviour; who knew what the subtleties of biochemistry-the quantum mechanics of the specific organic molecules involved-contributed to the nature of human consciousness? Copying the abstract neural topology wasn't enough. Sure, the jewel could pass the fatuous Turing test-no outside observer could tell it from a human-but that didn't prove that *being* a jewel felt the same as *being* human.

Eva asked, “Does that mean you'll never switch? You'll have your jewel removed? You'll let yourself *die* when your brain starts to rot?”

“Maybe,” I said. “Better to die at ninety or a hundred than kill myself at thirty, and have some machine marching around, taking my place, pretending to be me.”

“How do you know *I* haven't switched?” she asked, provocatively. “How do you know that I'm not just

‘pretending to be me’?”

“I know you haven't switched,” I said, smugly. “I just *know*.”

“How? I'd look the same. I'd talk the same. I'd act the same in every way. People are switching younger, these days. *So how do you know I haven't?*”

I turned onto my side towards her, and gazed into her eyes. “Telepathy. Magic. The communion of souls.”

My twelve-year-old self started snickering, but by then I knew exactly how to drive him away.

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At nineteen, although I was studying finance, I took an undergraduate philosophy unit. The Philosophy Department, however, apparently had nothing to say about the Ndoli Device, more commonly known as

“the jewel.” (Ndoli had in fact called it “the *dual*,” but the accidental, homophonic nick-name had stuck.) They talked about Plato and Descartes and Marx, they talked about St. Augustine and—when feeling particularly modern and adventurous—Sartre, but if they'd heard of Godel, Turing, Hamsun or Kim, they refused to admit it. Out of sheer frustration, in an essay on Descartes I suggested that the notion of human

consciousness as “software” that could be “implemented” equally well on an organic brain or an optical crystal was in fact a throwback to Cartesian dualism: for “software” read “soul.” My tutor superimposed a neat, diagonal, luminous red line over each paragraph that dealt with this idea, and wrote in the margin (in vertical, bold-face, 20-point Times, with a contemptuous 2 Hertz flash): IRRELEVANT!

I quit philosophy and enrolled in a unit of optical crystal engineering for non-specialists. I learnt a lot of solid-state quantum mechanics. I learnt a lot of fascinating mathematics. I learnt that a neural net is a device used only for solving problems that are far too hard to be *understood*. A sufficiently flexible neural net can be configured by feedback to mimic almost any

system-to produce the same patterns of output from the same patterns of input-but achieving this sheds no light whatsoever on the nature of the system being emulated.

“Understanding,” the lecturer told us, “is an overrated concept. Nobody really *understands* how a fertilized egg turns into a human. What should we do? Stop having children until ontogenesis can be described by a set of differential equations?”

I had to concede that she had a point there.

It was clear to me by then that nobody had the answers I craved-and I was hardly likely to come up with them myself; my intellectual skills were, at best, mediocre. It came down to a simple choice: I could waste time fretting about the mysteries of consciousness, or, like everybody else, I could stop worrying and get on with my life.

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When I married Daphne, at twenty-three, Eva was a distant memory, and so was any thought of the communion of souls. Daphne was thirty-one, an executive in the merchant bank that had hired me during my PhD, and everyone agreed that the marriage would benefit my career. What she got out of it, I was never quite sure. Maybe she actually liked me. We had an agreeable sex life, and we comforted each other when we were down, the way any kind-hearted person would comfort an animal in distress.

Daphne hadn't switched. She put it off, month after month, inventing ever more ludicrous excuses, and I teased her as if I'd never had reservations of my own.

“I'm afraid,” she confessed one night. “What if I die when it happens-what if all that's left is a robot, a puppet, a *thing*? I don't want to *die*.”

Talk like that made me squirm, but I hid my feelings. “Suppose you had a stroke,” I said glibly, “which destroyed a small part of your brain. Suppose

the doctors implanted a machine to take over the functions which that damaged region had performed. Would you still be 'yourself'?"

"Of course."

"Then if they did it twice, or ten times, or a thousand times-"

"That doesn't necessarily follow."

"Oh? At what magic percentage, then, would you stop being 'you'?"

She glared at me. "All the old clichéd arguments-"

"Fault them, then, if they're so old and clichéd."

She started to cry. "I don't have to. Fuck you! I'm scared to death, and you don't give a shit!"

I took her in my arms. "Sssh. I'm sorry. But everyone does it sooner or later. You mustn't be afraid. I'm here. I love you." The words might have been a recording, triggered automatically by the sight of her tears.

"Will you do it? With me?"

I went cold. "What?"

"Have the operation, on the same day? Switch when I switch?"

Lots of couples did that. Like my parents. Sometimes, no doubt, it was a matter of love, commitment, sharing. Other times, I'm sure, it was more a matter of neither partner wishing to be an unswitched person living with a jewel-head.

I was silent for a while, then I said, "Sure."

In the months that followed, all of Daphne's fears-which I'd mocked as "childish" and

“superstitious”-rapidly began to make perfect sense, and my own “rational” arguments came to sound abstract and hollow. I backed out at the last minute; I refused the anaesthetic, and fled the hospital.

Daphne went ahead, not knowing I had abandoned her.

I never saw her again. I couldn't face her; I quit my job and left town for a year, sickened by my cowardice and betrayal-but at the same time euphoric that I had *escaped*.

She brought a suit against me, but then dropped it a few days later, and agreed, through her lawyers, to an uncomplicated divorce. Before the divorce came through, she sent me a brief letter:

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There was nothing to fear, after all. I'm exactly the person I've always been. Putting it off was insane; now that I've taken the leap of faith, I couldn't be more at ease.

Your loving robot wife,

Daphne

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By the time I was twenty-eight, almost everyone I knew had switched. All my friends from university had done it. Colleagues at my new job, as young as twenty-one, had done it. Eva, I heard through a friend of a friend, had done it six years before.

The longer I delayed, the harder the decision became. I could talk to a thousand people who had switched, I could grill my closest friends for hours about their childhood memories and their most private thoughts, but however compelling their words, I knew that the Ndoli Device had spent decades buried in their heads, learning to fake exactly this kind of behaviour.

Of course, I always acknowledged that it was equally impossible to be *certain* that even another *unswitched* person had an inner life in any way the same as my own-but it didn't seem unreasonable to be more inclined to give the benefit of the doubt to people whose skulls hadn't yet been scraped out with a curette.

I drifted apart from my friends, I stopped searching for a lover. I took to working at home (I put in longer hours and my productivity rose, so the company didn't mind at all). I couldn't bear to be with people

whose humanity I doubted.

I wasn't by any means unique. Once I started looking, I found dozens of organisations exclusively for people who hadn't switched, ranging from a social club that might as easily have been for divorcees, to a paranoid, paramilitary "resistance front," who thought they were living out *Invasion of the Body Snatchers*. Even the members of the social club, though, struck me as extremely maladjusted; many of them shared my concerns, almost precisely, but my own ideas from other lips sounded obsessive and ill-

conceived. I was briefly involved with an unswitched woman in her early forties, but all we ever talked about was our fear of switching. It was masochistic, it was suffocating, it was insane.

I decided to seek psychiatric help, but I couldn't bring myself to see a therapist who had switched. When I finally found one who hadn't, she tried to talk me into helping her blow up a power station, to let THEM

know who was boss.

I'd lie awake for hours every night, trying to convince myself, one way or the other, but the longer I dwelt upon the issues, the more tenuous and elusive they became. Who was "I," anyway? What did it mean that "I" was "still alive," when my personality was utterly different from that of two decades before? My earlier selves were as good as dead-I remembered them no more clearly than I remembered contemporary acquaintances-yet this loss caused me only the slightest discomfort. Maybe the destruction of my organic brain would be the merest hiccup, compared to all the changes that I'd been through in my life so far.

Or maybe not. Maybe it would be exactly like dying.

Sometimes I'd end up weeping and trembling, terrified and desperately lonely, unable to comprehend-and yet unable to cease contemplating-the dizzying prospect of my own nonexistence. At other times, I'd simply grow "healthily" sick of the whole tedious subject. Sometimes I felt certain that the nature of the jewel's inner life was the most important question humanity could ever confront. At other times, my qualms seemed fey and laughable. Every day, hundreds of thousands of people switched, and the world apparently went on as always; surely that fact carried more weight than any abstruse philosophical argument?

Finally, I made an appointment for the operation. I thought, what is there to lose? Sixty more years of uncertainty and paranoia? If the human race was replacing itself with clockwork automata, I was better off dead; I lacked the blind conviction to join the psychotic underground-who, in any case, were tolerated by the authorities only so long as they remained ineffectual. On the other hand, if all my fears were unfounded-if my sense of identity could

survive the switch as easily as it had already survived such traumas as sleeping and waking, the constant death of brain cells, growth, experience, learning and forgetting-then I would gain not only eternal life, but an end to my doubts and my alienation.

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I was shopping for food one Sunday morning, two months before the operation was scheduled to take place, flicking through the images of an on-line grocery catalogue, when a mouth-watering shot of the latest variety of apple caught my fancy. I decided to order half a dozen. I didn't, though. Instead, I hit the key which displayed the next item. My mistake, I knew, was easily remedied; a single keystroke could take me back to the apples. The screen showed pears, oranges, grapefruit. I tried to look down to see what my clumsy fingers were up to, but my eyes remained fixed on the screen.

I panicked. I wanted to leap to my feet, but my legs would not obey me. I tried to cry out, but I couldn't make a sound. I didn't feel injured, I didn't feel weak. Was I paralysed? Brain-damaged? I could still *feel* my fingers on the keypad, the soles of my feet on the carpet, my back against the chair.

I watched myself order pineapples. I felt myself rise, stretch, and walk calmly from the room. In the kitchen, I drank a glass of water. I should have been trembling, choking, breathless; the cool liquid flowed smoothly down my throat, and I didn't spill a drop.

I could only think of one explanation: *I had switched*. Spontaneously. The jewel had taken over, while my brain was still alive; all my wildest paranoid fears had come true.

While my body went ahead with an ordinary Sunday morning, I was lost in a claustrophobic delirium of helplessness. The fact that everything I did was exactly what I had planned to do gave me no comfort. I caught a train to the beach, I swam for half an hour; I might as well have been running amok with an axe, or crawling naked down the street, painted with my own

excrement and howling like a wolf. *I'd lost control.* My body had turned into a living strait-jacket, and I couldn't struggle, I couldn't scream, I couldn't even close my eyes. I saw my reflection, faintly, in a window on the train, and I couldn't begin to guess what the mind that ruled that bland, tranquil face was thinking.

Swimming was like some sense-enhanced, holographic nightmare; I was a volitionless object, and the perfect familiarity of the signals from my body only made the experience more horribly *wrong*. My arms had no right to the lazy rhythm of their strokes; I wanted to thrash about like a drowning man, I wanted to show the world my distress.

It was only when I lay down on the beach and closed my eyes that I began to think rationally about my situation.

The switch *couldn't* happen “spontaneously.” The idea was absurd. Millions of nerve fibres had to be severed and spliced, by an army of tiny surgical robots which weren't even present in my brain-which weren't due to be injected for another two months. Without deliberate intervention, the Ndoli Device was utterly passive, unable to do anything but *eavesdrop*. No failure of the jewel or the teacher could possibly take control of my body away from my organic brain.

Clearly, there had been a malfunction-but my first guess had been wrong, absolutely wrong.

I wish I could have done *something*, when the understanding hit me. I should have curled up, moaning and screaming, ripping the hair from my scalp, raking my flesh with my fingernails. Instead, I lay flat on my back in the dazzling sunshine. There was an itch behind my right knee, but I was, apparently, far too lazy to scratch it.

Oh, I ought to have managed, at the very least, a good, solid bout of hysterical laughter, when I realised that I was the jewel.

The teacher had malfunctioned; it was no longer keeping me aligned with the organic brain. I hadn't suddenly become powerless; I had *always been* powerless. My will to act upon “my” body, upon the world, had *always*

gone straight into a vacuum, and it was only because I had been ceaselessly manipulated, “corrected” by the teacher, that my desires had ever coincided with the actions that seemed to be mine.

There are a million questions I could ponder, a million ironies I could savour, but *I mustn't*. I need to focus all my energy in one direction. My time is running out.

When I enter hospital and the switch takes place, if the nerve impulses I transmit to the body are not exactly in agreement with those from the organic brain, the flaw in the teacher will be discovered. *And rectified*. The organic brain has nothing to fear; *his* continuity will be safeguarded, treated as precious, sacrosanct. There will be no question as to which of us will be allowed to prevail. *I* will be made to conform, once again. *I* will be “corrected.” *I* will be murdered.

Perhaps it is absurd to be afraid. Looked at one way, I've been murdered every microsecond for the last twenty-eight years. Looked at another way, I've only existed for the seven weeks that have now passed since the teacher failed, and the notion of my separate identity came to mean anything at all—and in one more week this aberration, this nightmare, will be over. Two months of misery; why should I begrudge losing that, when I'm on the verge of inheriting eternity? Except that it won't be *I* who inherits it, since that two months of misery is all that defines me.

The permutations of intellectual interpretation are endless, but ultimately, I can only act upon my desperate will to survive. I don't *feel* like an aberration, a disposable glitch. How can I possibly hope to survive? I must conform-of my own free will. I must choose to make myself *appear* identical to that which they would force me to become.

After twenty-eight years, surely I am still close enough to him to carry off the deception. If I study every clue that reaches me through our shared senses, surely I can put myself in his place, forget, temporarily, the revelation of my separateness, and force myself back into synch.

It won't be easy. He met a woman on the beach, the day I came into being. Her name is Cathy. They've slept together three times, and he thinks he

loves her. Or at least, he's said it to her face, he's whispered it to her while she slept, he's written it, true or false, into his diary.

I feel nothing for her. She's a nice enough person, I'm sure, but I hardly know her. Preoccupied with my plight, I've paid scant attention to her conversation, and the act of sex was, for me, little more than a distasteful piece of involuntary voyeurism. Since I realised what was at stake, I've *tried* to succumb to the same emotions as my alter ego, but how can I love her when communication between us is impossible, when she doesn't even know *I* exist?

If she rules his thoughts night and day, but is nothing but a dangerous obstacle to me, how can I hope to achieve the flawless imitation that will enable me to escape death?

He's sleeping now, so I must sleep. I listen to his heartbeat, his slow breathing, and try to achieve a tranquillity consonant with these rhythms. For a moment, I am discouraged. Even my *dreams* will be different; our divergence is ineradicable, my goal is laughable, ludicrous, pathetic. Every nerve impulse, for a week? My fear of detection and my attempts to conceal it will, unavoidably, distort my responses; this knot of lies and panic will be impossible to hide.

Yet as I drift towards sleep, I find myself believing that I *will* succeed. I *must*. I dream for a while-a confusion of images, both strange and mundane, ending with a grain of salt passing through the eye of a needle-then I tumble, without fear, into dreamless oblivion.

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I stare up at the white ceiling, giddy and confused, trying to rid myself of the nagging conviction that there's something I *must* not think about.

Then I clench my fist gingerly, rejoice at this miracle, and remember.

Up until the last minute, I thought he was going to back out again-but he didn't. Cathy talked him through his fears. Cathy, after all, has switched, and he loves her more than he's ever loved anyone before.

So, our roles are reversed now. This body is *his* strait-jacket, now...

I am drenched in sweat. *This is hopeless, impossible.* I can't read his mind, I can't guess what he's trying to do. Should I move, lie still, call out, keep silent? Even if the computer monitoring us is programmed to

ignore a few trivial discrepancies, as soon as he notices that his body won't carry out his will, he'll panic just as I did, and I'll have no chance at all of making the right guesses. Would *he* be sweating, now?

Would *his* breathing be constricted, like this? *No.* I've been awake for just thirty seconds, and already I have betrayed myself. An optical-fibre cable trails from under my right ear to a panel on the wall.

Somewhere, alarm bells must be sounding.

If I made a run for it, what would they do? Use force? I'm a citizen, aren't I? Jewel-heads have had full legal rights for decades; the surgeons and engineers can't do anything to me without my consent. I try to recall the clauses on the waiver he signed, but he hardly gave it a second glance. I tug at the cable that holds me prisoner, but it's firmly anchored, at both ends.

When the door swings open, for a moment I think I'm going to fall to pieces, but from somewhere I find the strength to compose myself. It's my neurologist, Dr Prem. He smiles and says, "How are you feeling?"

Not too bad?"

I nod dumbly.

"The biggest shock, for most people, is that they don't feel different at all! For a while you'll think, 'It can't be this simple! It can't be this easy! It can't be this *normal*!' But you'll soon come to accept that *it is*. And life will go

on, unchanged.” He beams, taps my shoulder paternally, then turns and departs.

Hours pass. *What are they waiting for?* The evidence must be conclusive by now. Perhaps there are procedures to go through, legal and technical experts to be consulted, ethics committees to be assembled to deliberate on my fate. I'm soaked in perspiration, trembling uncontrollably. I grab the cable several times and yank with all my strength, but it seems fixed in concrete at one end, and bolted to my skull at the other.

An orderly brings me a meal. “Cheer up,” he says. “Visiting time soon.”

Afterwards, he brings me a bedpan, but I'm too nervous even to piss.

Cathy frowns when she sees me. “What's wrong?”

I shrug and smile, shivering, wondering why I'm even trying to go through with the charade. “Nothing. I just ... feel a bit sick, that's all.”

She takes my hand, then bends and kisses me on the lips. In spite of everything, I find myself instantly aroused. Still leaning over me, she smiles and says, “It's over now, okay? There's nothing left to be afraid of. You're a little shook up, but you know in your heart you're still who you've always been. And I love you.”

I nod. We make small talk. She leaves. I whisper to myself, hysterically, “I'm still who I've always been.

I'm still who I've always been.”

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Yesterday, they scraped my skull clean, and inserted my new, non-sentient, space-filling mock-brain.

I feel calmer now than I have for a long time, and I think at last I've pieced together an explanation for my survival.

Why do they deactivate the teacher, for the week between the switch and the destruction of the brain?

Well, they can hardly keep it running while the brain is being trashed-but why an entire week? To reassure people that the jewel, unsupervised, can still stay in synch; to persuade them that the life the

jewel is going to live will be exactly the life that the organic brain "would have lived"-whatever that could mean.

Why, then, only for a week? Why not a month, or a year? Because the jewel *cannot* stay in synch for that long-not because of any flaw, but for precisely the reason that makes it worth using in the first place.

The jewel is immortal. The brain is decaying. The jewel's imitation of the brain leaves out-deliberately-the fact that *real* neurons *die*. Without the teacher working to contrive, in effect, an identical deterioration of the jewel, small discrepancies must eventually arise. A fraction of a second's difference in responding to a stimulus is enough to arouse suspicion, and-as I know too well-from that moment on, the process of divergence is irreversible.

No doubt, a team of pioneering neurologists sat huddled around a computer screen, fifty years ago, and contemplated a graph of the probability of this radical divergence, versus time. How would they have chosen *one week*? What probability would have been acceptable? A tenth of a percent? A hundredth? A thousandth? However safe they decided to be, it's hard to imagine them choosing a value low enough to make the phenomenon rare on a global scale, once a quarter of a million people were being switched every day.

In any given hospital, it might happen only once a decade, or once a century, but every institution would still need to have a policy for dealing with the eventuality.

What would their choices be?

They could honour their contractual obligations and turn the teacher on again, erasing their satisfied customer, and giving the traumatised organic brain the chance to rant about its ordeal to the media and the legal profession.

Or, they could quietly erase the computer records of the discrepancy, and calmly remove the only witness.

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So, this is it. Eternity.

I'll need transplants in fifty or sixty years' time, and eventually a whole new body, but that prospect shouldn't worry me- *I* can't die on the operating table. In a thousand years or so, I'll need extra hardware tacked on to cope with my memory storage requirements, but I'm sure the process will be uneventful. On a time scale of millions of years, the structure of the jewel is subject to cosmic-ray damage, but error-free transcription to a fresh crystal at regular intervals will circumvent that problem.

In theory, at least, I'm now guaranteed either a seat at the Big Crunch, or participation in the heat death of the universe.

I ditched Cathy, of course. I might have learnt to like her, but she made me nervous, and I was thoroughly sick of feeling that I had to play a role.

As for the man who claimed that he loved her-the man who spent the last week of his life helpless, terrified, suffocated by the knowledge of his impending death-I can't yet decide how I feel. I ought to be able to empathise-considering that I once expected to suffer the very same fate myself-yet somehow he simply isn't *real* to me. I know my brain was modelled on his-giving him a kind of causal primacy-but in spite of that, I think of him now as a pale, insubstantial shadow.

After all, I have no way of knowing if his sense of himself, his deepest inner life, his experience of *being*,

was in any way comparable to my own.

About the Author

Greg Egan (1961-) obtained a BS in mathematics from the University of Western Australia and has worked as a computer programmer in medical research. The latest of his fascinating novels, *Teranesia*, appeared in 1999 from HarperCollins

